

VOLUME I - PROBLEM RESEARCH / DOCUMENT 3 OF 3

Field Requirements Specification

Functional, Performance, Interface and Acceptance Requirements for a Low-Visibility Haulage System

Document ID	NMDC-LVH-PR-03
Volume	I - Problem Research
Programme	Safe & Continuous Haulage under Low-Visibility Monsoon Conditions
Site of Application	Bailadila Iron Ore Mines - Kirandul & Bacheli Complexes, Dantewada, Chhattisgarh
Classification	Internal Research / Solution Design Input
Revision	v1.0
Issue Date	20 September 2026

ABSTRACT

This document converts the problem statement and constraint register of Documents 1 and 2 into a structured, numbered and testable requirements specification. It derives the performance envelope from first principles - stopping distance on a wet 1-in-16 gradient determines required detection range, and available lane width determines required positional accuracy - then specifies functional capability, human-machine interface behaviour, communications, control room function, environmental qualification, and a monsoon-scheduled acceptance test regime. Requirements are prioritised M (mandatory), S (should) and C (could) to support phased procurement.

Prepared as part of a solution-design research package addressing dense-fog and low-visibility haul road operations in large-scale mechanised open cast iron ore mines. Figures relating to NMDC production, statutory provisions and commercial systems are compiled from public sources; site-specific values marked as indicative require validation against NMDC operational records before use in a formal proposal.

Contents

- 1. Purpose and Structure**
- 2. Operating Modes and Visibility Classes**
- 3. Derivation of the Performance Envelope**
 - 3.1 Stopping distance analysis
 - 3.2 Required detection range
 - 3.3 Required positional accuracy
 - 3.4 Required latency budget
- 4. Functional Requirements**
 - 4.1 Positioning and lane guidance
 - 4.2 Hazard detection
 - 4.3 Cooperative awareness
 - 4.4 Alerting and intervention
- 5. Human-Machine Interface Requirements**
- 6. Communications Requirements**
- 7. Control Room and Decision Support Requirements**
- 8. Environmental and Qualification Requirements**
- 9. Integration, Data and Security Requirements**
- 10. Reliability, Maintainability and Availability**
- 11. Regulatory Requirements**
- 12. Acceptance Criteria and Test Regime**
- 13. Key Performance Indicators**

1. Purpose and Structure

This specification defines what a low-visibility haulage system for the Bailadila complexes must do, how well it must do it, and how that will be proved. It is written to be usable directly as the technical annexure of a tender or as the requirements baseline of an internal development programme.

Requirement identifiers take the form FR-xx (functional), PR-xx (performance), HR-xx (human-machine interface), CR-xx (communications), OR-xx (control room), ER-xx (environmental), IR-xx (integration), RR-xx (reliability), GR-xx (regulatory) and AR-xx (acceptance).

Priority is one of:

- ? **M - Mandatory.** The system is not fit for purpose without it.
- ? **S - Should.** Strongly desirable; omission requires justification and a compensating measure.
- ? **C - Could.** Valuable; candidate for a later phase.

Each requirement traces back to the constraint register (C-xx) of Document 2 where applicable.

2. Operating Modes and Visibility Classes

The system shall replace the present binary go/no-go response with a graded operating regime. Visibility classes are defined by measured meteorological optical range (MOR), not by subjective assessment.

Class	MOR band	Designation	Intended operating posture
V0	> 200 m	Clear	Normal operation. System runs in background, logging and monitoring only.
V1	50 - 200 m	Light haze	Normal operation with system advisory active. Speed unrestricted beyond normal limits.
V2	20 - 50 m	Moderate fog	Assisted operation. Advisory speed guidance active. Forward hazard alerting active. Light vehicle movement restricted to escorted or system-equipped only.
V3	5 - 20 m	Dense fog	Guided operation. Lane guidance primary reference. Reduced advisory speed. One-way or slot-managed movement on critical sections under control room sequencing.
V4	< 5 m	Severe fog	Minimum-speed guided movement on designated instrumented routes only, or controlled suspension. Applies where the system's validated performance supports it; otherwise suspension.

The present state of practice is effectively V0-V1 normal, and suspension at V2 or V3. The programme objective is to convert V2 and V3 into productive assisted and guided operating classes, and to make V4 a controlled, instrumented decision rather than a blanket stoppage. This is where the production recovery is found.

FR-01 (M). The system shall determine the prevailing visibility class automatically from calibrated visibility sensing, and shall not rely on subjective human assessment for mode determination. (*traces C-01*)

FR-02 (M). Visibility class shall be determined **per zone**, with zones defined by elevation band and haul road segment, reflecting the strong vertical stratification and patchiness of Bailadila fog. (*traces Vol. I Doc. 1 Sec. 3.2*)

FR-03 (M). Transition between classes shall apply hysteresis and a minimum dwell time to prevent rapid mode oscillation during transient advection fog.

FR-04 (S). The system shall provide a short-horizon visibility nowcast, with a target horizon of 15-30 minutes, to support anticipatory fleet sequencing rather than reactive stoppage.

3. Derivation of the Performance Envelope

Performance requirements are derived rather than asserted, so that they can be re-derived if fleet or road parameters change.

3.1 Stopping distance analysis

The stopping distance of a vehicle on a gradient is:

$$d_{\text{stop}} = v t_{\text{pr}} + v^2 / (2 g * (\mu_{\text{eff}} + G))$$

where v is speed (m/s), t_{pr} is perception-reaction time (s), g is 9.81 m/s^2 , μ_{eff} is the effective deceleration coefficient achievable by the vehicle on the prevailing surface, and G is the gradient as a decimal (negative on descent).

Parameter selection for a loaded rear-dump truck on a wet Bailadila ramp:

Parameter	Value adopted	Basis
t_{pr}	1.5 s	Alerted operator, elevated workload in fog; conservative relative to the 1.0 s often used for expectant drivers
μ_{eff}	0.25	Loaded haul truck, wet compacted ore-fines surface. Deliberately conservative: haul trucks are retarder-braked, mass is high, and a wet high-fines surface offers markedly less friction than sealed road
G	-0.0625	1 in 16 descent, the DGMS maximum gradient

Table 3.1 - Parameters adopted for stopping distance derivation. These are indicative and shall be replaced by measured braking performance for the actual Bailadila fleet during requirement finalisation.

Resulting stopping distances on a 1-in-16 wet descent:

Speed (km/h)	Speed (m/s)	Reaction distance (m)	Braking distance (m)	Total stopping distance (m)
10	2.78	4.2	4.1	8.3
15	4.17	6.3	9.3	15.6
20	5.56	8.3	16.5	24.8
25	6.94	10.4	25.8	36.2
30	8.33	12.5	37.2	49.7
40	11.11	16.7	66.1	82.8

Table 3.2 - Derived stopping distance, loaded truck, wet surface, 1-in-16 descent, $\mu_{\text{eff}} = 0.25$, $t_{\text{pr}} = 1.5 \text{ s}$.

Two conclusions follow immediately.

First, **at MOR 3-5 m the safe speed is effectively walking pace.** Visual operation cannot exceed roughly 5 km/h without the stopping distance exceeding the sight distance. This is exactly what is observed in practice and explains the magnitude of the production loss.

Second, a **10-15 m proximity envelope supports about 12-15 km/h at best**. To support a useful haul speed of 20-25 km/h the system must detect hazards at 25-40 m; to support 30 km/h it must detect at 50 m. This single calculation invalidates the existing generation of short-range proximity devices for this application and sets the headline sensing requirement.

3.2 Required detection range

A safety margin factor of 1.5 is applied to the derived stopping distance to account for parameter uncertainty, surface variability, detection latency and the need to give the operator a decision rather than an emergency.

PR-01 (M). The system shall detect a haul-truck-sized target on the haul road at a minimum range of **80 m**, with a design goal of 120 m, under conditions of MOR down to 5 m. *(supports 30 km/h with margin; traces C-03)*

PR-02 (M). The system shall detect a light-vehicle-sized target at a minimum range of **50 m** under MOR down to 5 m.

PR-03 (M). The system shall detect a person-sized target at a minimum range of **25 m** under MOR down to 5 m, and throughout the full 360 degree blind zone envelope of the host machine at short range.

PR-04 (S). The system shall detect a static non-cooperative obstruction of minimum dimension 0.5 m in the travel path at a minimum range of 40 m under MOR down to 5 m. *(addresses rockfall, dropped material, washout edge)*

PR-05 (M). Detection performance shall be specified as a probability of detection of at least 0.95 per target per second within the stated range, with a false alarm rate not exceeding the figure specified in HR-06.

PR-06 (M). Range accuracy shall be better than +/- 0.5 m or 2 per cent of range, whichever is greater; range rate accuracy better than +/- 0.3 m/s.

PR-07 (S). Azimuth accuracy shall be sufficient to discriminate an in-lane target from an adjacent-lane target at 50 m, implying an angular resolution of the order of 3 degrees or better at that range for the prevailing road width.

3.3 Required positional accuracy

Lane width available to a haul truck on a two-way Bailadila haul road is approximately the running surface width less berms and clearance, divided by two. For a large rear-dump truck the lateral margin between the vehicle envelope and the lane boundary is of the order of 1-2 m.

PR-08 (M). Absolute horizontal positional accuracy shall be **better than 0.30 m (95 per cent)** in open-sky conditions, degrading to no worse than 1.0 m (95 per cent) under partial sky obstruction typical of pit and ramp geometry. *(traces C-04, C-05)*

PR-09 (M). Positional solution shall be continuously available with no gap exceeding 2 seconds, achieved by inertial and odometric coasting through GNSS outages. *(traces C-05)*

PR-10 (M). Heading accuracy shall be better than 1.0 degree, achieved by dual-antenna GNSS or equivalent inertial means rather than magnetic heading. *(traces C-06 - magnetic heading is unreliable in high-grade haematite)*

PR-11 (M). The positioning solution shall provide a real-time integrity and confidence figure, and the human-machine interface shall degrade gracefully and visibly when confidence falls below the level required for lane guidance. *(traces C-17 - prevents over-trust)*

3.4 Required latency budget

End-to-end latency from the physical occurrence of a hazard to a perceptible operator alert:

Segment	Allocation
Sensor acquisition and signal processing	60 ms
Fusion, tracking and association	40 ms
Threat assessment and alert decision	30 ms
Interface rendering (visual and audible)	70 ms
Total system latency budget	200 ms

PR-12 (M). End-to-end latency from hazard occurrence to operator-perceptible alert shall not exceed **200 ms** at the 95th percentile.

PR-13 (M). Vehicle-to-vehicle message latency shall not exceed **100 ms** at the 95th percentile, and shall not depend on a server round trip or on cellular infrastructure availability. *(traces C-11)*

PR-14 (M). Cooperative position update rate shall be at least **10 Hz**.

4. Functional Requirements

4.1 Positioning and lane guidance

FR-05 (M). The system shall maintain a continuous estimate of host vehicle position, velocity, heading and attitude referenced to the mine survey datum.

FR-06 (M). The system shall hold a digital model of the active haul road network - centreline, edges, lane assignment, gradient, curvature, speed zones and restricted areas - and shall present the vehicle's position relative to that model.

FR-07 (M). The digital road model shall be **updatable from the field within one working shift** without vendor involvement, and shall carry a validity timestamp visible to the control room. *(traces C-13 - haul roads change continuously)*

FR-08 (M). The system shall provide continuous lateral deviation guidance relative to the assigned lane, with escalating indication as the vehicle approaches the lane boundary and a distinct indication for berm proximity.

FR-09 (S). The system shall provide a forward path preview along the road model - upcoming curve direction and severity, gradient change, junction, tipping point - at a look-ahead distance scaled to current speed.

FR-10 (M). The system shall detect and alert on departure from the assigned lane and on approach to a geofenced restricted area, dump edge or exclusion zone.

4.2 Hazard detection

FR-11 (M). The system shall detect **non-cooperative** hazards - objects, vehicles and persons carrying no transponder - using sensing modalities whose performance is substantially unaffected by fog. *(traces C-01, C-02; see Volume III Document 1)*

FR-12 (M). Forward sensing coverage shall span the vehicle's travel path and both adjacent lanes to the ranges of PR-01 to PR-04.

FR-13 (M). The system shall provide 360 degree short-range coverage of the host machine's blind zones, with particular attention to the rear and the near-side quarters of a rear-dump truck.

FR-14 (M). The system shall classify detected targets to at least the categories: large vehicle, light vehicle, person, static obstruction, unknown. Classification confidence shall be exposed to the alerting logic so that a low-confidence detection is treated conservatively rather than discarded.

FR-15 (M). The system shall track targets over time, maintaining identity, velocity and predicted trajectory, and shall not treat each detection as an independent event. *(essential to FR-19 trajectory-aware alerting)*

FR-16 (S). The system shall fuse cooperative (V2V-derived) and non-cooperative (sensed) tracks into a single target list, resolving duplicates, and shall indicate to the operator which contacts are cooperative and which are sensed. *(traces C-16, F4 - operators must know what the system cannot see)*

FR-17 (S). The system shall detect degradation of its own sensing capability - aperture fouling, obstruction, calibration drift, sensor fault - and shall declare a degraded mode rather than continuing to present a complete-looking picture. *(traces C-07, F2, F3)*

4.3 Cooperative awareness

FR-18 (M). Every equipped vehicle shall broadcast, at not less than 10 Hz, a basic status message containing at minimum: identity, class, position, position confidence, velocity vector, heading, and operating state (loaded, empty, stationary, reversing, tipping, broken down).

FR-19 (M). Each vehicle shall receive and process the broadcasts of all vehicles within communications range and shall compute a threat assessment for each based on relative position, closing rate, predicted trajectory and the road model - **not on straight-line distance alone**. *(traces C-16, F1 - distance-only logic is the principal cause of alarm fatigue)*

FR-20 (M). The cooperative layer shall function **directly between vehicles**, without dependence on infrastructure or backhaul availability. *(traces C-11, F6)*

FR-21 (S). Personnel on foot in operating areas shall be equipped with a compatible personal device providing the same cooperative presence, with a battery life of not less than one full shift plus margin, and an unambiguous low-battery and fault indication.

FR-22 (S). The system shall support broadcast of static and temporary hazard markers - a broken-down machine, an active road maintenance area, a washout, a blast exclusion zone - placed by a field user or by the control room, and shall present these to approaching vehicles.

4.4 Alerting and intervention

FR-23 (M). Alerting shall be graded across at least three levels - awareness, caution and warning - determined by time-to-collision or time-to-lane-departure rather than by raw distance.

FR-24 (M). Alert thresholds shall adapt to the prevailing visibility class, vehicle speed, load state, gradient and road geometry.

FR-25 (S). The system shall provide advisory speed guidance consistent with the prevailing visibility class, the system's currently validated detection range, and the derived stopping distance of the host machine.

FR-26 (C). The system may provide automatic intervention - speed limiting or braking - subject to a separate DGMS approval pathway. Intervention shall be architecturally separable from advisory function so that advisory deployment is not delayed by intervention approval. *(traces C-22, GR-03)*

FR-27 (M). The operator shall at all times be able to determine the system's current health, mode and effective sensing range from the interface without navigating menus. *(traces C-17, F5)*

5. Human-Machine Interface Requirements

HR-01 (M). The primary display shall convey the tactical situation **spatially and pre-attentively**. A plan or perspective view of the host vehicle, the road corridor and surrounding contacts shall be interpretable in a single glance of under 1.5 seconds. *(traces C-15)*

HR-02 (M). The interface shall be predominantly **non-textual**. Where text is unavoidable it shall be configurable across the languages in use on site, and no safety-critical information shall depend on reading text. *(traces C-18)*

HR-03 (M). Colour shall not be the sole carrier of safety-critical information; shape, position, size and motion shall be used redundantly, accommodating colour vision deficiency and display degradation.

HR-04 (M). Audible alerts shall be distinguishable in a cab ambient of up to 85 dB(A) with hearing protection in use, and shall be directional or otherwise encode bearing where technically feasible. Alert tones shall be distinct from existing plant and reverse alarms. *(traces Doc. 2 Sec. 2.4 - habituation)*

HR-05 (S). A haptic channel - seat or control-mounted - should be provided for the highest alert level, as the modality least susceptible to masking in a high-noise, high-visual-load environment.

HR-06 (M). The **false or nuisance alert rate shall not exceed 1 per operating hour per vehicle** averaged over a shift in normal congested operation, and shall be a formal, measured acceptance criterion. *(traces C-16, F1 - this is the single most important human factors requirement in the specification)*

HR-07 (M). Display luminance shall auto-adjust across full daylight glare to night conditions, with a night mode that preserves operator dark adaptation.

HR-08 (M). The display shall be positioned and sized so that reading it does not require the operator to move their head substantially from the forward view, and shall be mountable without obstructing existing controls, mirrors or the operator's field of view.

HR-09 (M). The interface shall be operable with gloved hands and shall require no interaction during normal driving.

HR-10 (S). Operators shall be able to acknowledge an alert and to report a nuisance alert with a single physical action, with the report logged for tuning. *(builds the feedback loop that keeps HR-06 satisfied over time)*

HR-11 (M). The system shall never present a blank, ambiguous or apparently normal display when it is not functioning. Loss of function shall be unmistakable. *(traces F3)*

6. Communications Requirements

CR-01 (M). The vehicle-to-vehicle link shall operate on a modality unaffected by fog and shall meet PR-13 and PR-14.

CR-02 (M). Effective V2V range shall be not less than **500 m line of sight**, with graceful degradation in the non-line-of-sight conditions produced by ridge and pit geometry.

CR-03 (M). The V2V link shall support the peak vehicle density anticipated in congested areas - loading faces, crusher tips, workshop approaches - without message loss materially degrading threat assessment. A design density of at least 60 vehicles within mutual range shall be assumed.

CR-04 (M). Messages shall be authenticated to prevent spoofing of position or identity, and the system shall reject implausible messages.

CR-05 (S). Vehicle-to-infrastructure communication shall be provided for road-model updates, hazard markers, dispatch instructions and telemetry upload, and shall tolerate intermittent connectivity with store-and-forward

behaviour. (*traces C-11*)

CR-06 (M). Loss of infrastructure connectivity shall not disable any safety-critical function. The vehicle shall remain fully protective on V2V and on-board sensing alone. (*traces F6*)

CR-07 (M). All radio equipment shall comply with the applicable Indian spectrum regulations, shall use licence-exempt or appropriately licensed bands, and shall be demonstrated not to interfere with existing site radio or blast initiation systems.

CR-08 (S). The system should coexist with, and where possible reuse, any existing site wireless network rather than requiring a wholly parallel RF estate.

7. Control Room and Decision Support Requirements

OR-01 (M). The control room shall have a real-time common operating picture showing all equipped vehicles and personnel on a mine plan, with position, class, state, speed and system health.

OR-02 (M). The picture shall be overlaid with per-zone visibility class, derived from the visibility sensing network of FR-01 and FR-02.

OR-03 (M). The system shall log all alerts, near-miss events, lane departures, speed exceedances, mode changes and system faults, with position and timestamp, in a form suitable for safety investigation and for DGMS reporting.

OR-04 (S). The control room shall be able to issue graded operating instructions - mode change, zone restriction, one-way or slot allocation on a section, hold or release - and these shall be delivered to affected vehicle displays.

OR-05 (S). The system shall provide decision support for the fog response: current and nowcast visibility by zone, current effective fleet productivity, and the projected production consequence of alternative responses, so that the decision to reduce or suspend is informed rather than reflexive. (*directly addresses levers 3 and 4 of Vol. I Doc. 1 Sec. 4.3*)

OR-06 (S). The system shall support re-sequencing and re-dispatch of the fleet on restart, to reduce the restart overhead t_r .

OR-07 (M). System health shall be visible at fleet level, with any vehicle whose safety function is degraded or failed clearly flagged. (*traces F3*)

OR-08 (C). A digital twin of the haul road network and fleet may be maintained for scenario analysis, route and schedule optimisation under fog conditions, and operator training.

OR-09 (M). The control room function shall be presented within, or tightly integrated with, the existing dispatch and fleet management environment rather than as an additional standalone console. (*traces C-14, F7*)

8. Environmental and Qualification Requirements

ER-01 (M). Externally mounted equipment shall be rated not less than IP67; equipment subject to pressure washing shall be IP69K. (*traces C-07*)

ER-02 (M). Equipment shall operate across an ambient range of -5 deg C to +65 deg C, and shall tolerate engine-bay and cab-roof soak temperatures where so mounted.

ER-03 (M). Equipment shall operate at sustained relative humidity to 95 per cent with condensation cycling. Internal electronics shall be conformally coated. (*traces C-08*)

ER-04 (M). Equipment shall be qualified to the shock and vibration levels of HEMM service. Rotating media and unsecured connectors are prohibited. *(traces C-09)*

ER-05 (M). Power input shall accept a wide-range DC supply around 24 V nominal with load-dump, reverse-polarity and transient protection to vehicular standards, and shall be galvanically isolated from machine networks.

ER-06 (M). Outdoor fixed infrastructure shall incorporate surge and lightning protection on power and antenna feeds. *(traces C-10)*

ER-07 (M). Any optical or radar aperture shall be positioned, shielded or actively cleaned to resist ore-mud accumulation, and its obscuration shall be self-detected per FR-17. *(traces C-07)*

ER-08 (M). Fixed roadside nodes shall be capable of autonomous solar-plus-battery operation with **not less than 7 days of autonomy at monsoon insolation levels**, and shall be relocatable by site personnel without specialist tooling. *(traces C-12, C-13, F8)*

ER-09 (M). All equipment shall be demonstrated electromagnetically compatible with existing site systems, with explicit verification against blast initiation systems.

ER-10 (S). RF and magnetic subsystems shall be validated **on the Bailadila site** in the presence of high-grade haematite masses, not solely in a generic test environment. *(traces C-06)*

9. Integration, Data and Security Requirements

IR-01 (M). The system shall expose open, documented interfaces for position, event, alert and health data, avoiding proprietary lock-in.

IR-02 (M). The system shall integrate with the existing fleet management, dispatch and weighbridge systems so that cycle, productivity and safety data reside in one coherent estate. *(traces C-14, F7)*

IR-03 (M). Time across all vehicles and infrastructure shall be synchronised to a common reference, with synchronisation accuracy adequate for the latency requirements of Section 3.4.

IR-04 (M). A written data governance policy shall be agreed **before deployment**, covering what operator-attributable data is collected, who may access it, for what purposes it may and may not be used, and retention periods. *(traces C-19, F9 - this is a deployment-critical requirement, not an administrative formality)*

IR-05 (M). Vehicle-side and infrastructure equipment shall be secured against unauthorised access, and firmware shall be updated only through an authenticated mechanism.

IR-06 (M). Equipment shall be tamper-evident and shall report disconnection, disablement or removal to the control room. *(traces C-24, F3)*

IR-07 (S). The system shall support phased rollout, operating correctly and safely on a partially equipped fleet, with explicit interface indication of the coverage limitation. *(traces F4)*

10. Reliability, Maintainability and Availability

RR-01 (M). Availability of the vehicle-borne safety function shall be not less than **99 per cent** of scheduled operating hours per vehicle, measured monthly.

RR-02 (M). The system shall perform a power-on self-test and continuous background self-test, and shall report health to the control room at least once per minute while operating.

RR-03 (M). Pre-shift verification of the system shall be integrated into the existing DGMS-mandated pre-operational checklist, with an unambiguous pass/fail indication that does not require interpretation. *(traces GR-02, F3)*

RR-04 (M). All units shall be field-replaceable by site maintenance personnel using documented procedures, with no requirement for vendor attendance for routine replacement. *(traces F12)*

RR-05 (M). Installation on any fleet machine shall be achievable within a single planned maintenance window using a documented, machine-specific harness kit. *(traces C-20, C-21)*

RR-06 (S). Calibration of multi-sensor extrinsics shall be monitored online, with automatic alert on drift beyond tolerance. *(traces F10)*

RR-07 (M). Mean time to repair for a failed vehicle unit shall not exceed one shift, with sufficient spares held on site.

11. Regulatory Requirements

GR-01 (M). The system shall comply with the Mines Act 1952 and the Metalliferous Mines Regulations 1961.

GR-02 (M). The system shall satisfy, and shall be documented as satisfying, the relevant provisions of DGMS Technical Circular 06 of 2020 (HEMM safety features including proximity warning devices) and DGMS Technical Circular 03 of 2024 (transport and tramming machinery safety, proximity detection systems, pre-operational checklists), and the audio-visual alarm provisions associated with Circular 08/2017 and Regulations 107/108. *(traces C-23)*

GR-03 (M). Any function taking control authority over the machine shall be subject to a defined DGMS approval, testing and commissioning pathway, engaged at programme inception. Advisory functions shall be architecturally separable so that their deployment is not gated by intervention approval. *(traces C-22)*

GR-04 (M). All radio equipment shall hold the applicable Indian type approval and spectrum authorisation.

GR-05 (S). Where the system alters established safe operating procedures - for example by permitting movement at a visibility at which haulage is presently suspended - the revised procedure shall be documented, risk-assessed and agreed with the statutory authority before adoption.

GR-06 (M). The event log of OR-03 shall be retained and made available in a form suitable for statutory accident investigation.

12. Acceptance Criteria and Test Regime

AR-01 (M). Acceptance shall comprise four stages: factory/laboratory test, controlled site trial in clear conditions, **instrumented site trial in measured fog conditions**, and a full-season operational evaluation.

AR-02 (M). Fog-condition acceptance testing shall be conducted with **calibrated, independent visibility measurement** (transmissometer or forward-scatter visibility meter) co-located with the test, so that performance is recorded against measured MOR and not against subjective description. *(traces C-26, F11)*

AR-03 (M). Detection performance requirements PR-01 to PR-07 shall be demonstrated against standardised targets at measured MOR values spanning classes V2, V3 and V4.

AR-04 (M). The false alert rate requirement HR-06 shall be measured over not less than 200 vehicle-operating-hours in normal congested operation, with operator-reported nuisance alerts as a cross-check.

AR-05 (M). Positional accuracy requirements PR-08 to PR-11 shall be verified against an independent survey-grade reference along representative ramp, switchback and pit-bottom sections, including sections of known GNSS shadowing.

AR-06 (M). Latency requirements PR-12 to PR-14 shall be measured instrumentally end to end, not estimated from component specifications.

AR-07 (M). Environmental qualification ER-01 to ER-09 shall be evidenced by test certificates; ER-10 shall be evidenced by on-site measurement at Bailadila.

AR-08 (M). Operational evaluation shall span **at least one complete monsoon season** before full-fleet commitment. (*traces F11, F12*)

AR-09 (S). A controlled comparison shall be established between equipped and unequipped sections or shifts to permit attribution of productivity and safety effects.

AR-10 (M). Operator acceptance shall be formally surveyed, covering trust, workload, perceived nuisance alert burden and willingness to rely on the system. A technically compliant system with poor operator acceptance shall not be accepted for full deployment. (*traces C-15 to C-18, F1, F5, F9*)

13. Key Performance Indicators

The programme shall be measured on outcome, not on installation count. Baseline values must be established from at least one prior monsoon season before deployment.

KPI	Definition	Direction
K1 - Fog stoppage hours	Haulage hours lost to visibility-related suspension, per monsoon month	Reduce
K2 - Stop-start event count	Discrete suspension and restart events per monsoon month	Reduce
K3 - Restart overhead	Mean time from release to restored fleet throughput	Reduce
K4 - Fog-condition haul speed	Mean loaded haul speed during V2/V3 conditions	Increase
K5 - Fog-condition cycle time	Mean haul cycle time during V2/V3 conditions	Reduce
K6 - Monsoon fleet utilisation	Effective fleet hours as a fraction of scheduled hours, monsoon months	Increase
K7 - Monsoon output	Tonnes despatched, June-October, normalised for fleet size and plant availability	Increase
K8 - Collision and near-miss rate	Reportable incidents and system-detected near-misses per million tonne-km, monsoon months	Reduce
K9 - Nuisance alert rate	Operator-reported and system-classified false alerts per vehicle-hour	Reduce; must satisfy HR-06
K10 - System availability	Per RR-01	Maintain above 99%
K11 - Operator trust index	Structured survey score, measured at intervals	Increase
K12 - Visibility nowcast skill	Verification score of the FR-04 nowcast against measured MOR	Increase

K9 and K11 are placed deliberately alongside the production and safety KPIs. Experience across the sector is that systems fail operationally long before they fail technically, and that the leading indicator of operational failure is a rising nuisance alert rate with a falling trust score. These should be monitored from the first week of the trial, not reviewed at its conclusion.